EXP 26

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clc;
clear;
close all;

%% 1. Define (7,4) Hamming Code
% Generator matrix
G = [1 0 0 0 1 1 0;
    0 1 0 0 1 0 1;
    0 0 1 0 0 1 1;
    0 0 0 1 1 1 1];

% Parity-check matrix
H = [1 1 0 1 1 0 0;
    1 0 1 1 0 1 0;
    0 1 1 1 0 0 1];

%% 2. Generate Data Bits
data = [1 0 1 1];

disp('Original Data Bits:');
disp(data);

%% 3. Encoding
% Codeword = data * G modulo 2
codeword = mod(data * G, 2);

disp('Encoded Codeword:');
disp(codeword);
```

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%% 4. Introduce a Single-Bit Error
received = codeword;

errorPosition = 3;  % Position of the bit error
received(errorPosition) = ...
    mod(received(errorPosition) + 1, 2);

disp('Received Codeword with Error:');
disp(received);

%% 5. Calculate Syndrome Vector
% Syndrome = received * H transpose modulo 2
syndrome = mod(received * H', 2);

disp('Syndrome Vector:');
disp(syndrome);

%% 6. Detect Error Position
if all(syndrome == 0)
    disp('No error detected.');
    corrected = received;
else
    % Find the column of H matching the syndrome
    errorFound = false;

    for i = 1:size(H, 2)
        if isequal(syndrome', H(:, i))
            errorPositionDetected = i;
            errorFound = true;
            break;
        end
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    end

    if errorFound
        fprintf('Error detected at bit position %d\n', ...
            errorPositionDetected);

        %% 7. Correct the Error
        corrected = received;
        corrected(errorPositionDetected) = ...
            mod(corrected(errorPositionDetected) + 1, 2);
    else
        disp('Error pattern is not a single-bit error.');
        corrected = received;
    end
end

%% 8. Display Corrected Codeword
disp('Corrected Codeword:');
disp(corrected);

%% 9. Extract Original Data Bits
decodedData = corrected(1:4);

disp('Recovered Data Bits:');
disp(decodedData);

%% 10. Verify Correction
if isequal(corrected, codeword)
    disp('SUCCESS: Single-bit error corrected.');
else
    disp('Correction unsuccessful.');
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end

%% 11. Plot Bit Sequences
figure('Name', 'Hamming Code Error Correction');

subplot(4,1,1);
stem(data, 'filled');
title('Original Data Bits');
xlabel('Bit Position');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;

subplot(4,1,2);
stem(codeword, 'filled');
title('Encoded 7-Bit Codeword');
xlabel('Bit Position');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;

subplot(4,1,3);
stem(received, 'filled');
title('Received Codeword with Single-Bit Error');
xlabel('Bit Position');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;

subplot(4,1,4);
stem(corrected, 'filled');
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```
title('Corrected Codeword');
xlabel('Bit Position');
ylabel('Bit Value');
ylim([-0.2 1.2]);
grid on;
```

